Yuru Sun

yuru.sun@monash.edu | https://yurusun.netlify.app/

Education

Monash University *Department of Econometrics and Business Statistics*

Doctor of Philosophy *Jan 2023 - present*

- My research interests falls in financial econometrics, Bayesian analysis and asset pricing. My current research project bridge the Bayesian non-parametric analysis to the empirical finance data, with particular focus on the price jumps.

University of Melbourne *Melbourne Business School*

Doctor of Philosophy *June 2022 - Jan 2023*

- I started my PhD at Melbourne Business School and then transferred to Monash University to continue my PhD degree.

Monash University *Monash Business School*

Bachelor of Commerce (1st Class Honors) *Feb 2017 - Dec 2020*

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

Research Experience

Jump Modeling with Dirichlet Process Mixture

Monash University & University of Melbourne *June 2022 - Present*

- **Abstract**: We propose a Bayesian nonparametric method using Dirichlet process mixture to model jumps. This new model encompasses existing parametric jump specification, but allows for extreme jumps to occur in either tail of the return distribution via infinite mixture model. The flexibility in our model specification allows for multi-modal jump size distribution. We investigate the robustness of our proposed model under different jump intensity and size specifications under simulations and provide comparisons about the model-driven option implied volatility smiles. We verify the practical use of our model with empirical data in jump detection, data-driven jump size density and implied volatility smiles.

Optimal probabilistic forecasts for risk management

Monash University *Mar 2020 - Mar 2023*

- **Abstract:** This paper explores the implications of producing forecast distributions that are optimized according to scoring rules that are relevant to financial risk management. We assess the predictive performance of optimal forecasts from potentially misspecified models for i) value-at-risk and expected shortfall predictions; and ii) prediction of the VIX volatility index for use in hedging strategies involving VIX futures. Our empirical results show that calibrating the predictive distribution using a score that rewards the accurate prediction of extreme returns improves the VaR and ES predictions. Tail-focused predictive distributions are also shown to yield better outcomes in hedging strategies using VIX futures.
- **Link to the paper**: https://arxiv.org/abs/2303.01651

Conference Presentations

The European Seminar on Bayesian Econometrics (ESOBE 2025)

Melbourne Business School, Australia *August, 2025*

- Accepted - conference will be held in 26-27 August, 2025.
- Poster presentation

The 8th International Conference on Econometrics and Statistics (EcoSta 2025)

Waseda University, Japan *August, 2025*

- Accepted - conference will be held in 21-23 August, 2025.
- Invited Session

World Congress of the Econometric Society

Seoul, South Korea *August, 2025*

Accepted - conference will be held in 18-22 August, 2025.

Annual Conference of the International Association for Applied Econometrics

University of Turin, Italy *June, 2025*

Accepted - conference will be held in 25-27 June, 2025.

The 19th International Symposium on Econometric Theory and Applications: SETA 2025

University of Macau, China *June, 2025*

Accepted - conference will be held in 1-3 June, 2025.

Econometric Society Australian Meetings (ESAM)
Monash University, Australia *December, 2024*

Time Series and Forecasting Symposium
University of Sydney, Australia *November, 2024*

- Poster presentation
- Best Poster Runner-up

Work Experience

Monash Business School *Melbourne, Australia*
Teaching Associate *Feb 2023 - present*

- High Dimensional Data Analysis (Head tutor)
- Data Analysis in Business (**Excellent teaching team**)
- Introductory Econometrics
- Business Statistics
- Business Forecasting

Monash Business School *Melbourne, Australia*
Research Assistant *Aug 2023 - Oct 2024*

- High frequency data analysis of bond data for the Thailand Bond Market Association. Evaluate bond price jumps under the influence of regulation changes.

Melbourne Business School *Melbourne, Australia*
Research Assistant *Aug 2022 - Jan 2022*

- Use Python to extract high-frequency data of each stock in S%P 500 list from database - Refinitiv
- Using R to finish the data cleaning job, e.g. Extract the five-minute price for every stock and aggregate high frequency return to daily level.

Honors & Awards

2025	**Prestigious International Conference Award**, Monash Business School
2025	**CFM/SoFiE Scholarship**, Capital Fund Management, The Society for Financial Econometrics
2024	**Best Poster Runner Up**, Time Series and Forecasting Symposium, University of Sydney
2023	**Co-funded Monash Graduate Scholarship**, Monash University
2022	**Faculty of Business and Economics Graduate Research Scholarship**, University of Melbourne
2021	**2020 Dean's Honor (First Class)**, Monash Business School
2020	**Econometrics Memorial Scholarship**, Department of Econometrics and Business Statistics, Monash
2020	**Meritorious Winner**, 2020 Interdisciplinary contest in Modelling

References available upon request.